Augmented Neural ODEs

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Latent ODEs for Irregularly-Sampled Time Series

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一、增强神经常微分方程（ANODE)

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回顾Neural ODE

$$\frac{\mathrm{d}\mathbf{h}(t)}{\mathrm{d}t} = \mathbf{f}(\mathbf{h}(t), t), \qquad \mathbf{h}(0) = \mathbf{x}$$

初值问题(IVP)

特征变换算子 $\phi_t(\mathbf{x}) = \mathbf{x} + \int_0^t \mathbf{f}(\mathbf{h}(\tau), \tau) d\tau$

该算子是微分同胚映射：
连续，逆映射连续，且一一映射

具有特点：

1 维度始终为R^d

2 连续时间演化

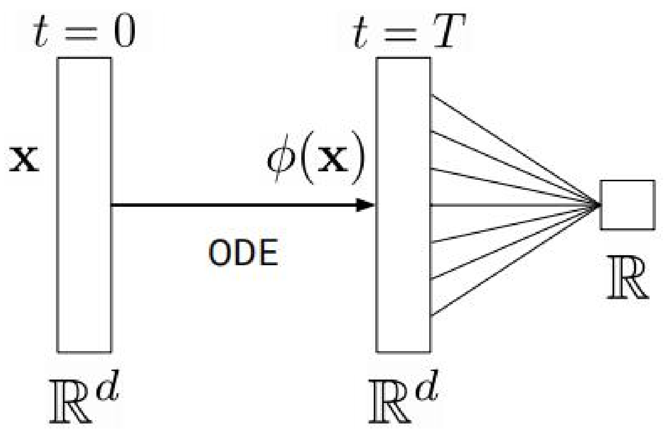
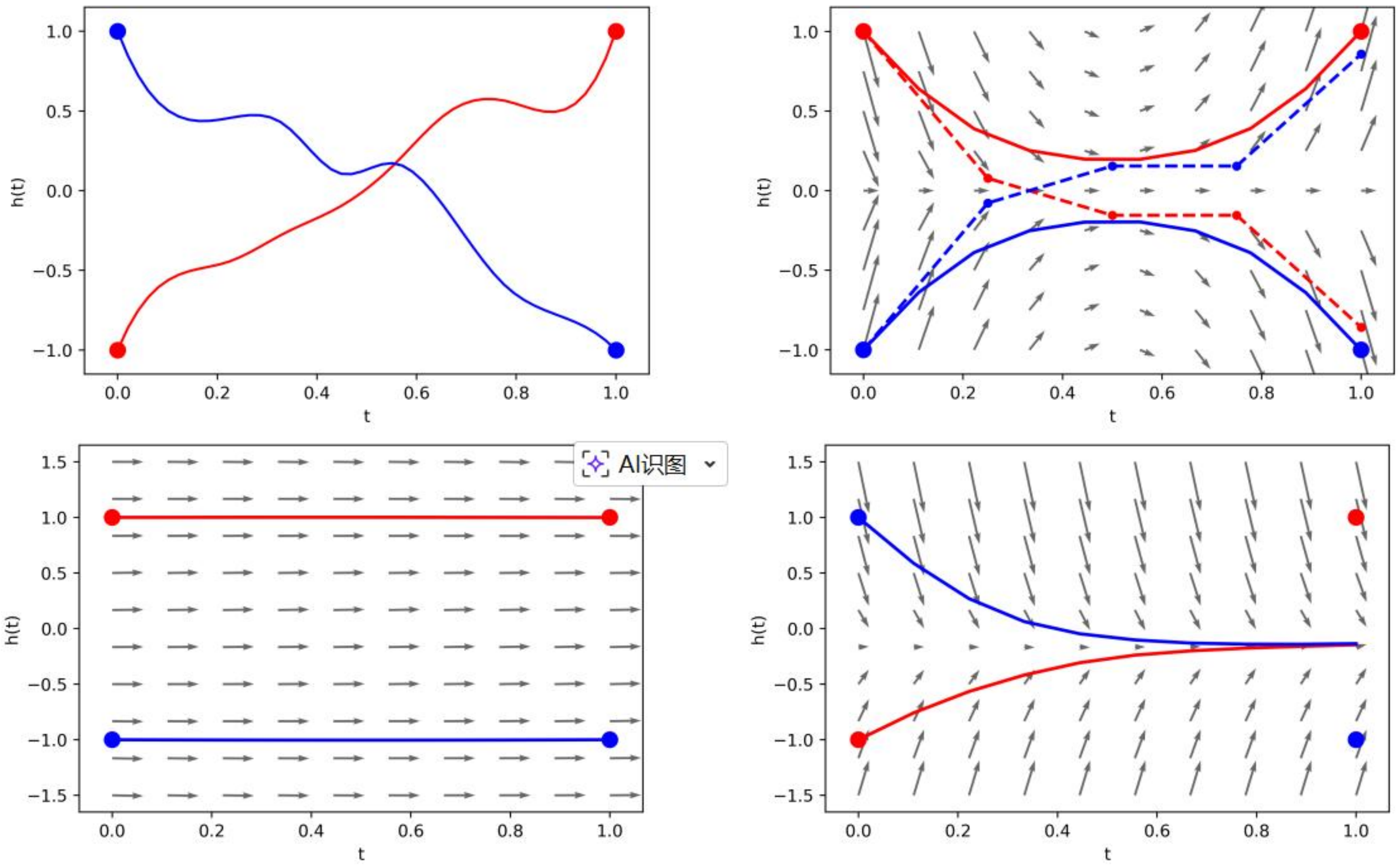


Figure 2: Diagram of Neural ODE architecture.

node遇到的问题

一维的一个简单示例

$$g_{1\mathrm{d}}(-1) = 1$$
$$g_{1\mathrm{d}}(1) = -1$$



node无法表示的一类函数

$$\begin{cases} g(\mathbf{x}) = -1 & \text{if } \|\mathbf{x}\| \leq r_1 \\ g(\mathbf{x}) = 1 & \text{if } r_2 \leq \|\mathbf{x}\| \leq r_3, \end{cases}$$

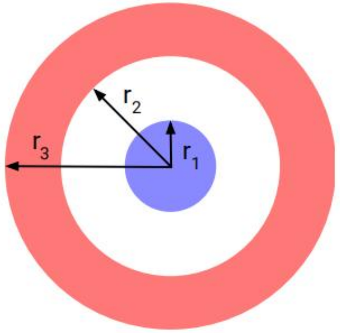
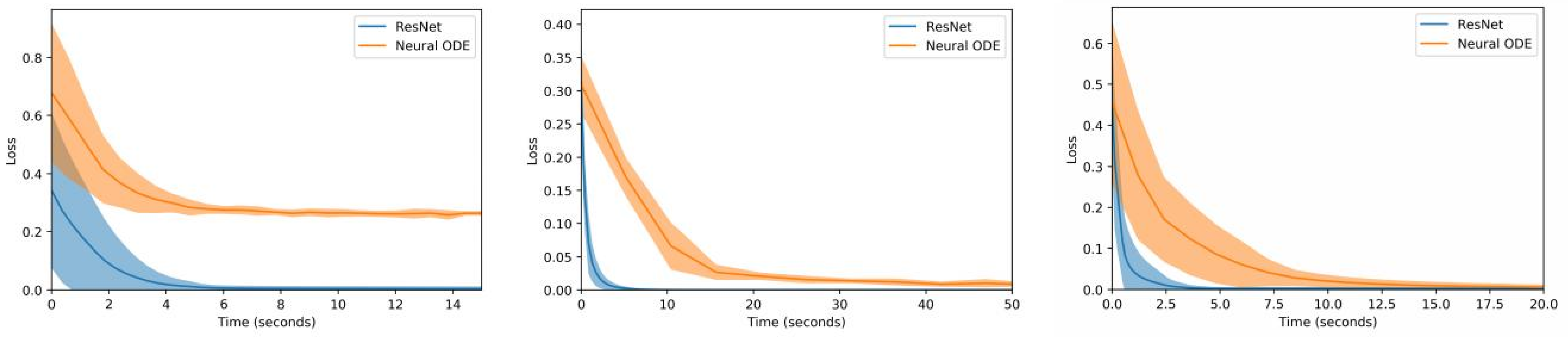


Figure 4: Diagram of $g(\mathbf{x})$ for $d = 2$.

node只能连续的变形输入空间



(a) $g(\mathbf{x})$ in $d = 1$ (b) $g(\mathbf{x})$ in $d = 2$ (c) Separable function in $d = 2$

理论 VS 实际

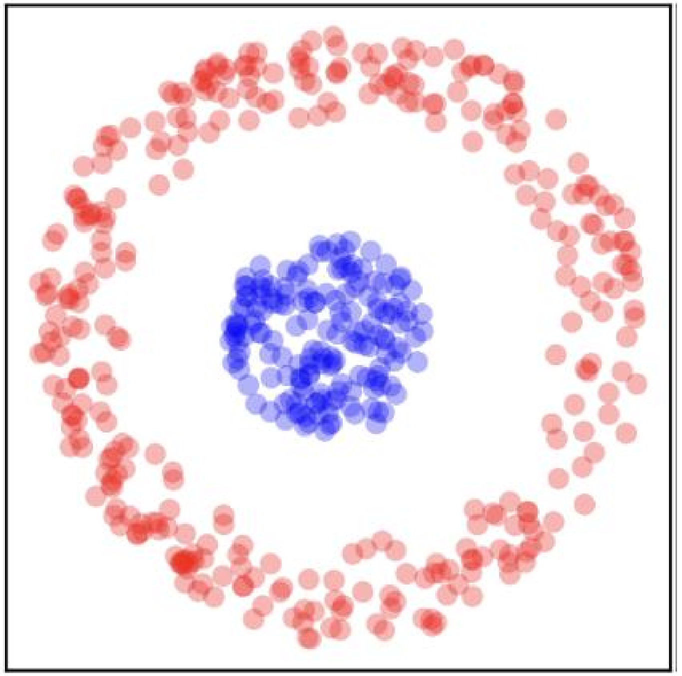
理论上：

蓝色点绝不可能出去

实际上：

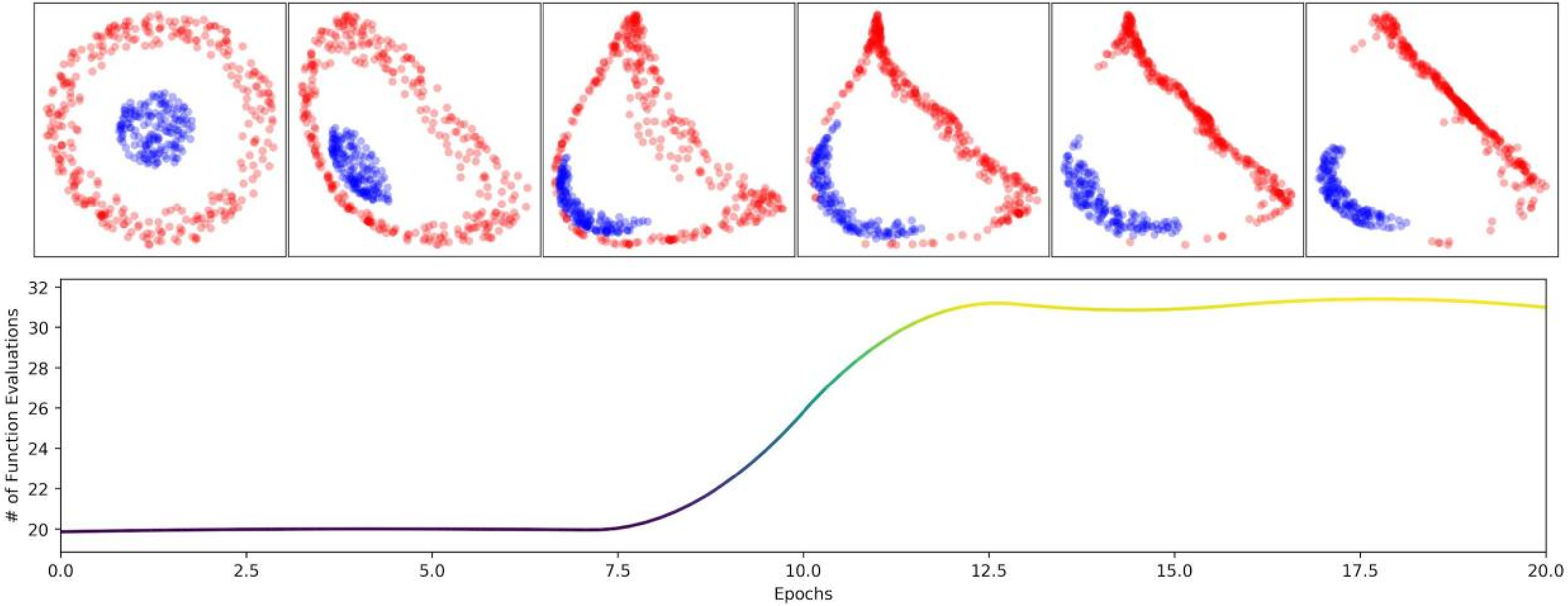
由于输入的采样点是离散的，并非密不透风

蓝色点可以从红点之间的缝隙硬挤出去



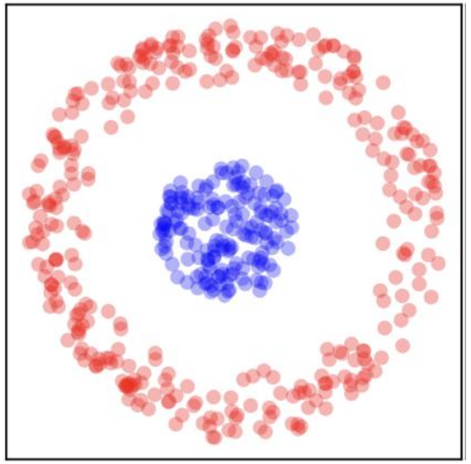
病态问题！！！

实验过程



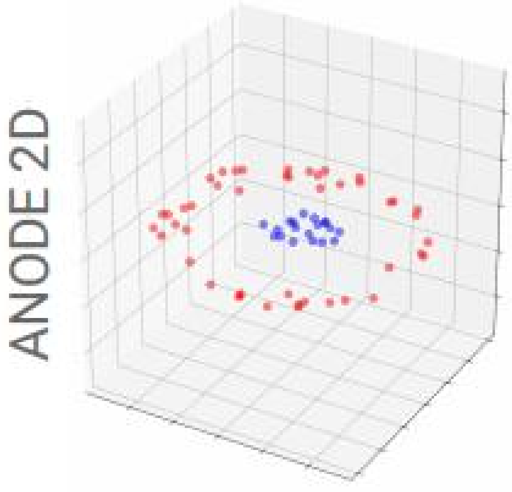
NFE 函数评估次数

Augmented Neural ODEs (ANODE)

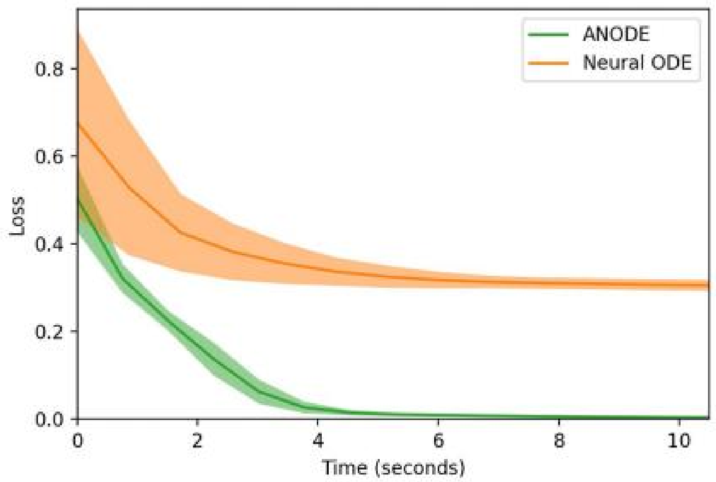


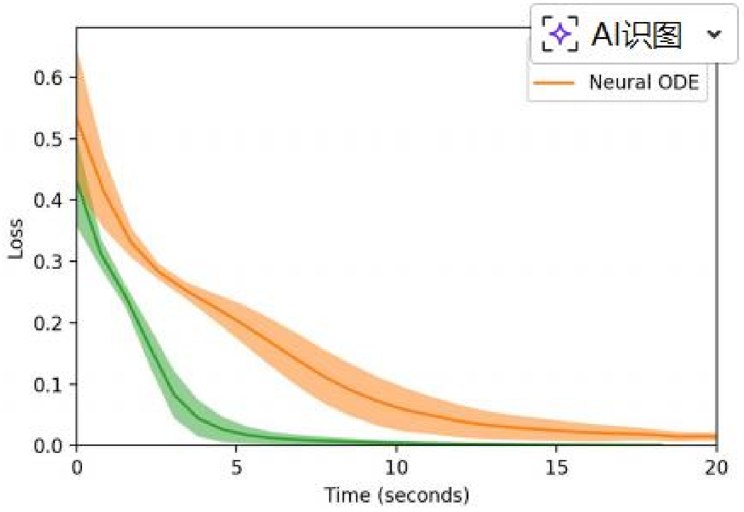
ANODE 强制将原始输入拼接上 p 维的零向量，将求解空间从 R^d 扩展到更高维度的 R^(d+p)

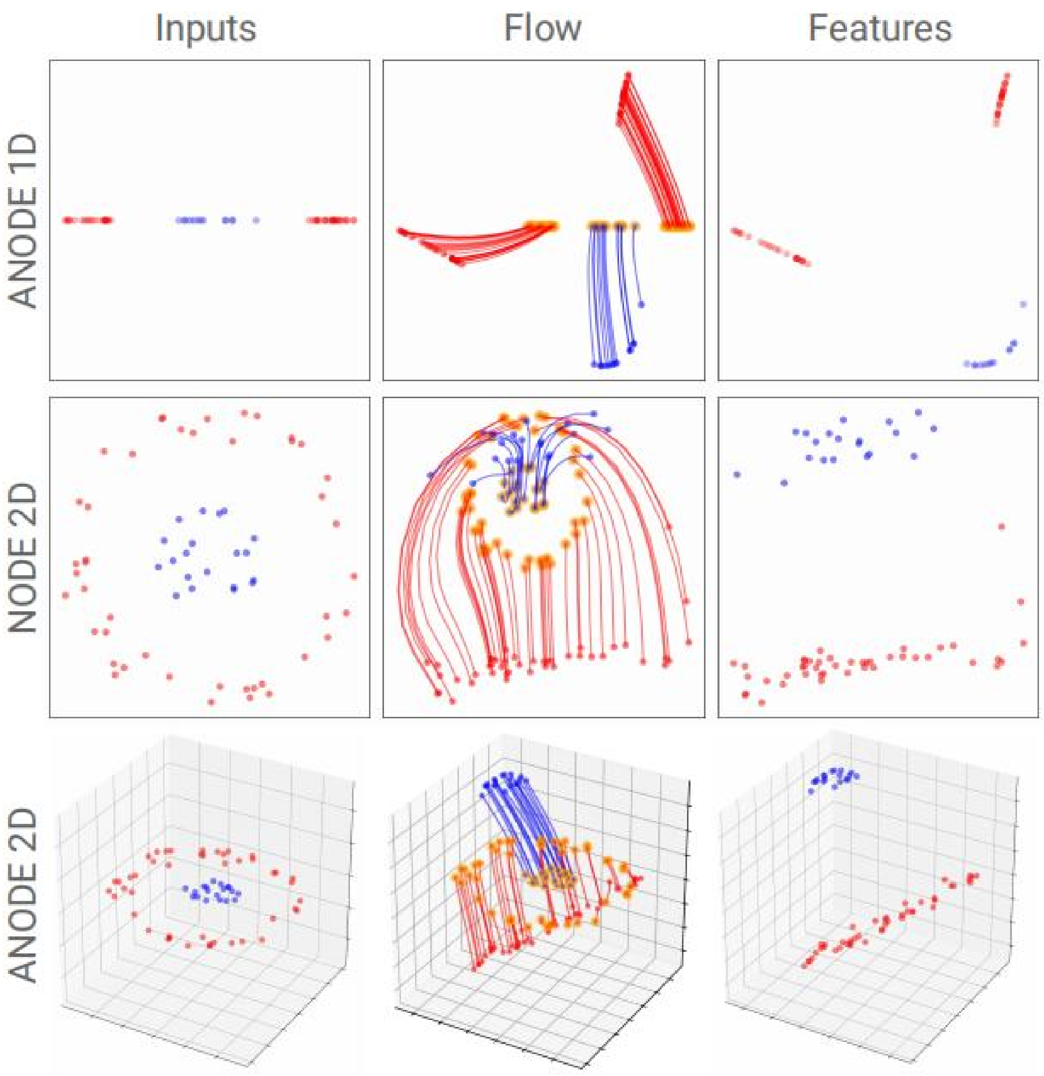
$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{bmatrix}\mathbf{h}(t)\\ \mathbf{a}(t)\end{bmatrix} = \mathbf{f}(\begin{bmatrix}\mathbf{h}(t)\\ \mathbf{a}(t)\end{bmatrix}, t), \qquad \begin{bmatrix}\mathbf{h}(0)\\ \mathbf{a}(0)\end{bmatrix} = \begin{bmatrix}\mathbf{x}\\ \mathbf{0}\end{bmatrix}$$



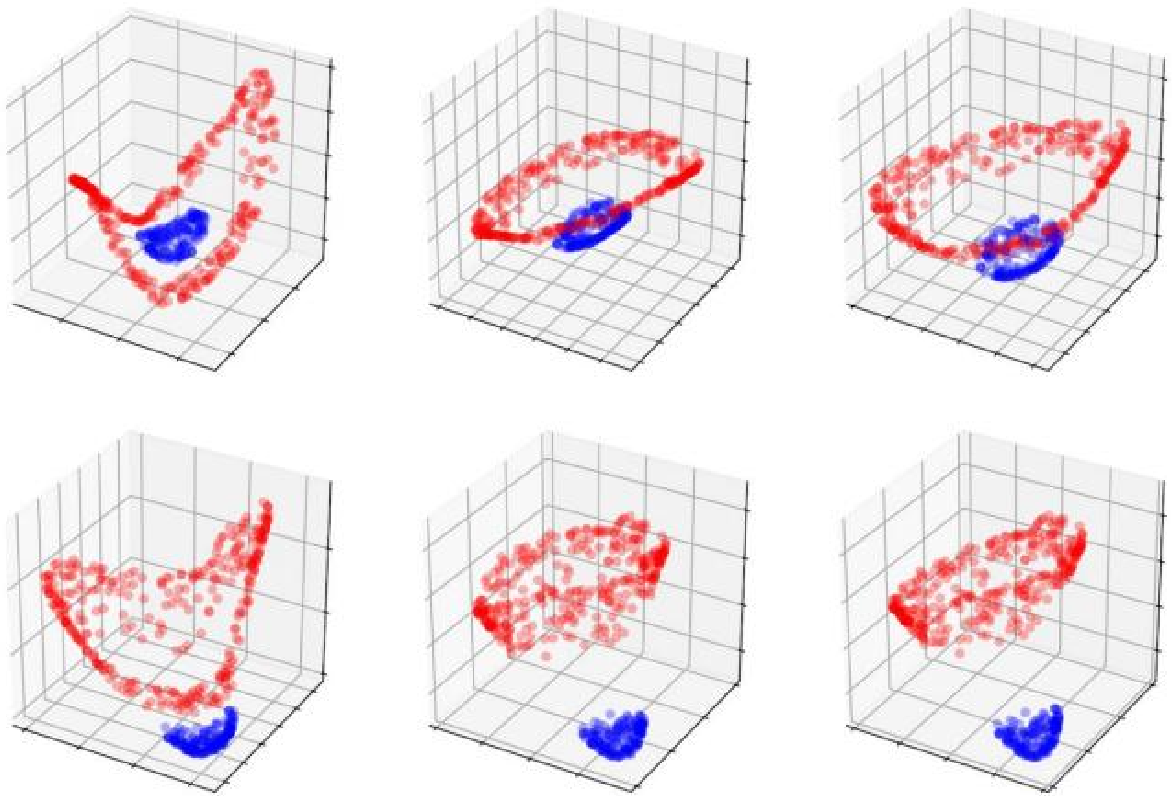
实验

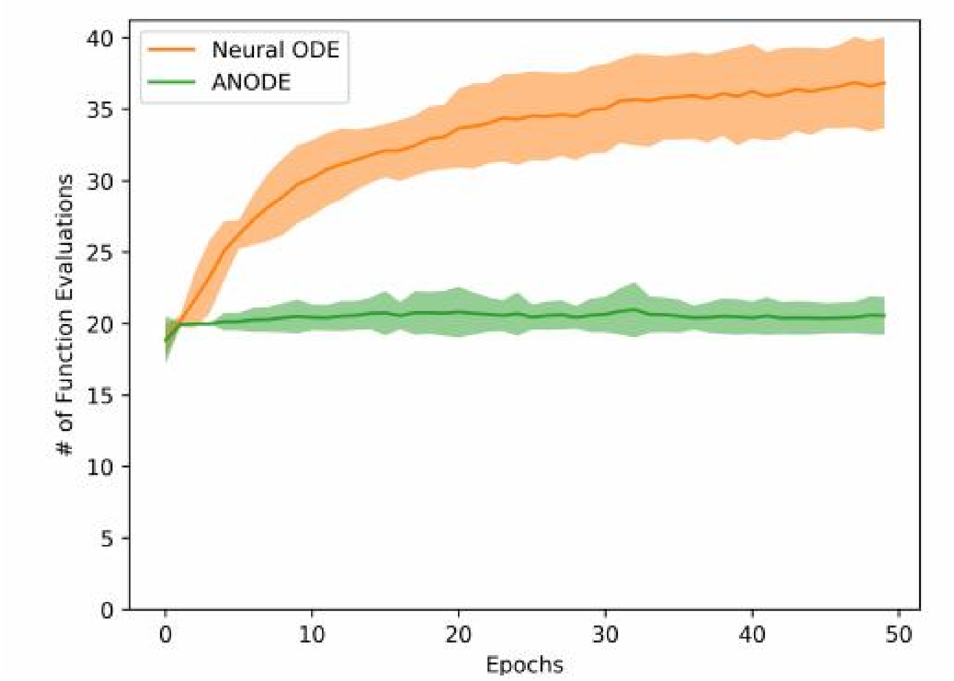

ANODE
Neural ODE
Loss
0.0
0.2
0.4
0.6
0.8
0
2
4
6
8
10
Time (seconds)


AI识图
Neural ODE
Loss
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0
5
10
15
20
Time (seconds)

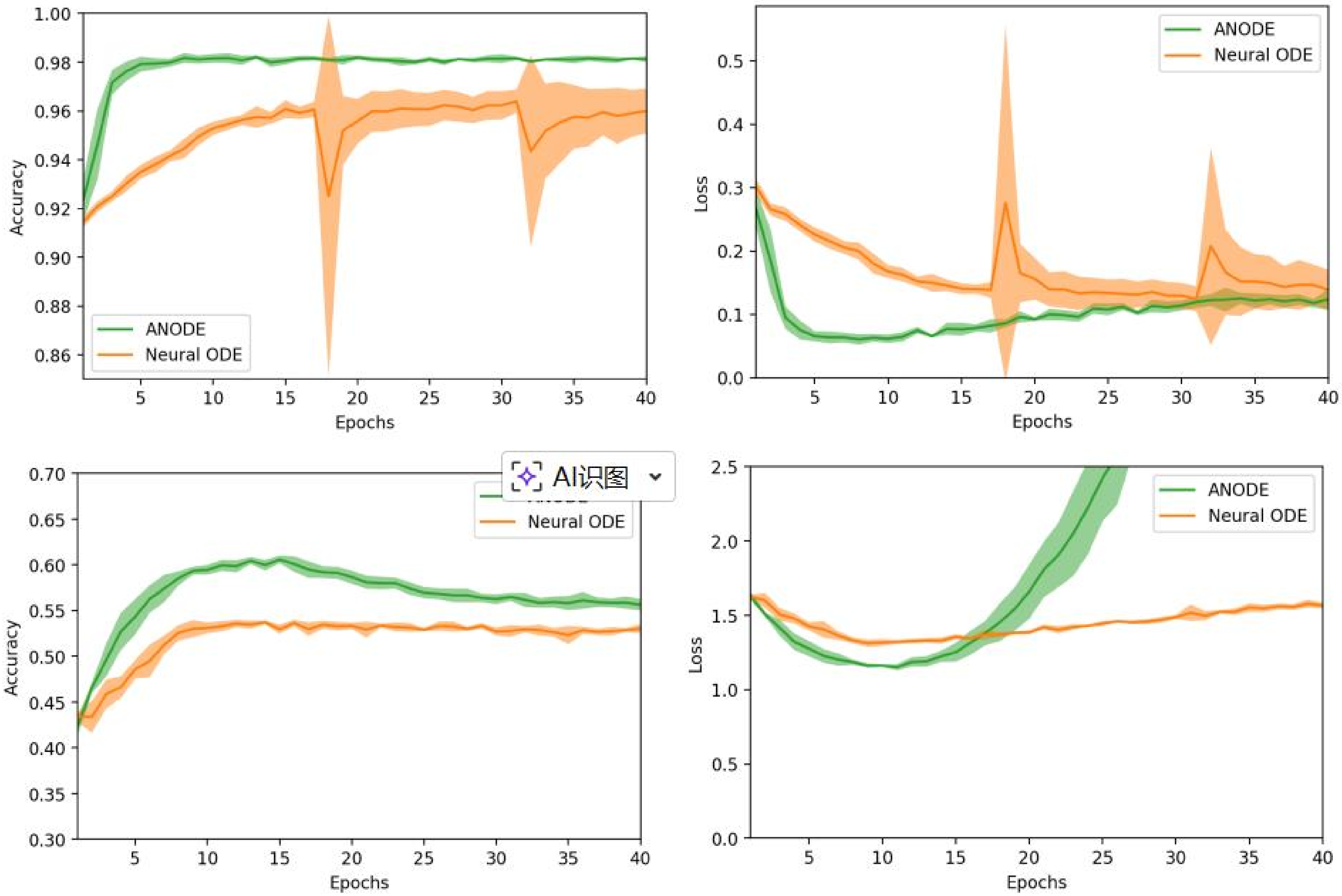

Inputs
Flow
Features
ANODE 1D
NODE 2D
ANODE 2D

实验




Neural ODE
ANODE
# of Function Evaluations
Epochs
0
5
10
15
20
25
30
35
40
50

泛化能力



优势

1.实现了更低的计算成本

2.更强的泛化能力

3.大多情况下更高的准确率

4.稳定性

5.扩展性

局限性

1.虽然anode比node快，但他仍然比ResNet慢

2.使用anode会改变空间的维度，这在某些实际情况中可能是不可取的

3.增加的维度d难以选择，当d选择不够好时，会导致模型表现更差，Loss和NFE更高

二、用于不规则采样时间序列的潜在微分方程

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潜空间不同演化路径

$$h_i = \mathrm{RNNCell}(h_{i-1}, \Delta_t, x_i) \qquad \Delta_t = t_i - t_{i-1}$$

$$h_i = \mathrm{RNNCell}(h_{i-1} \cdot \exp\{-\tau \Delta_t\}, x_i)$$

$$\frac{dh(t)}{dt} = -\tau h$$

$$\frac{dh(t)}{dt} = f_\theta(h(t), t) \quad \text{where} \quad h(t_0) = h_0$$

$$h_0, \ldots, h_N = \mathrm{ODESolve}(f_\theta, h_0, (t_0, \ldots, t_N))$$

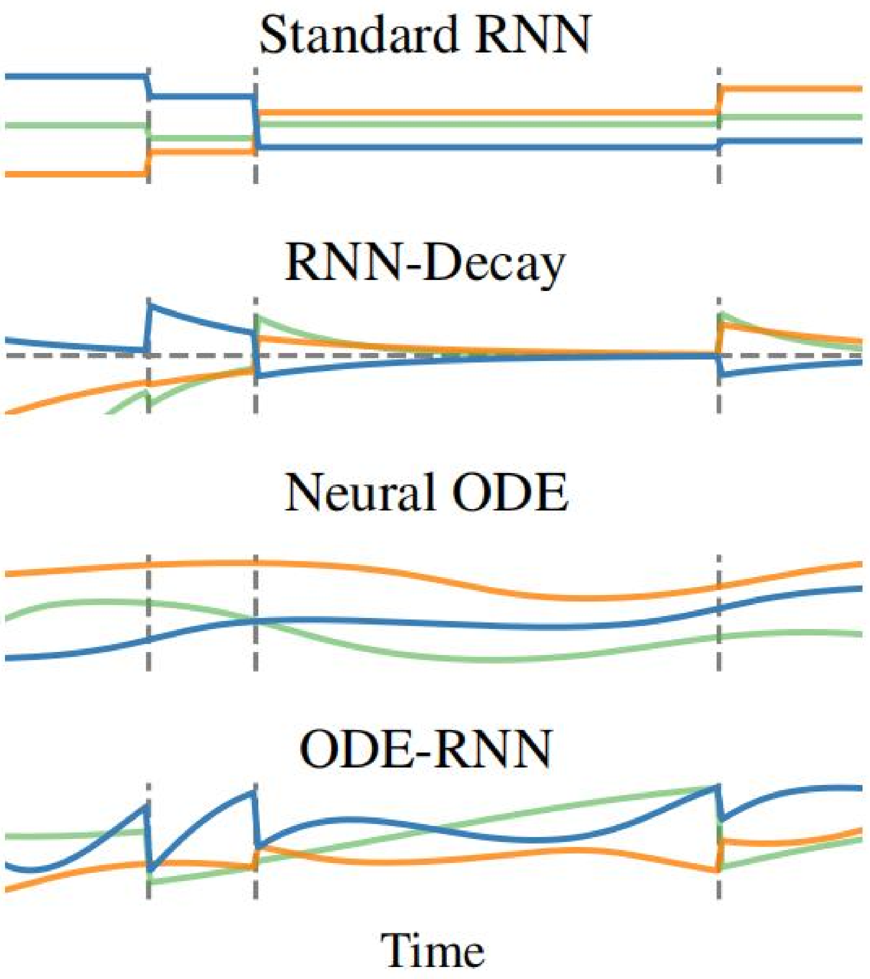


Figure 1: Hidden state trajectories.

ODE-RNN

阶段一：连续时间演化

$$h_i' = \text{ODESolve}(f_\theta, h_{i-1}, (t_{i-1}, t_i))$$

阶段二：离散状态跳跃

$$h_i = \text{RNNCell}(h_i', x_i)$$

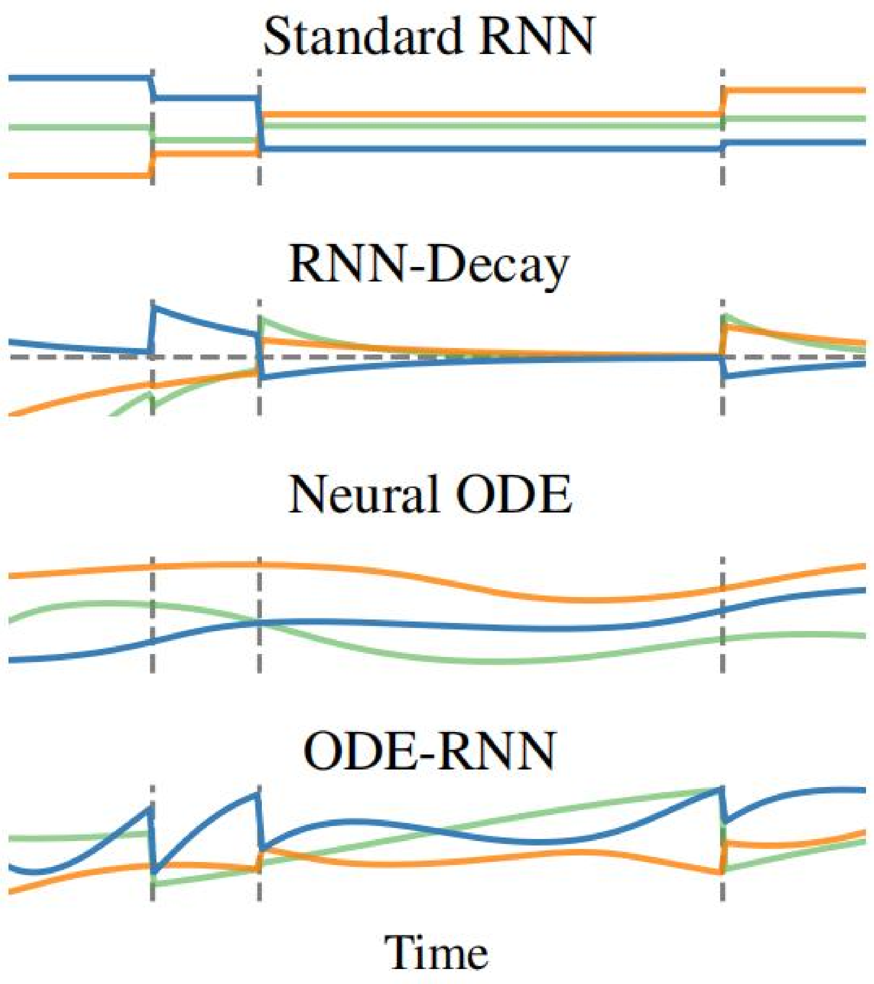


Figure 1: Hidden state trajectories.

ODE-RNN

Algorithm 1 The ODE-RNN. The only difference, highlighted in blue, from standard RNNs is that the pre-activations h' evolve according to an ODE between observations, instead of being fixed.

Input: Data points and their timestamps $\{(x_i, t_i)\}_{i=1..N}$
$h_0 = \mathbf{0}$
for i in 1, 2, …, N **do**
 $h'_i = \text{ODESolve}(f_\theta, h_{i-1}, (t_{i-1}, t_i))$ ▷ Solve ODE to get state at t_i
 $h_i = \text{RNNCell}(h'_i, x_i)$ ▷ Update hidden state given current observation x_i
end for
$o_i = \text{OutputNN}(h_i)$ for all $i = 1..N$
Return: $\{o_i\}_{i=1..N}; h_N$

Latent ODEs

$$z_0 \sim p(z_0)$$
$$z_0, z_1, \ldots, z_N = \mathrm{ODESolve}(f_\theta, z_0, (t_0, t_1, \ldots, t_N))$$
$$\text{each} \quad x_i \overset{indep.}{\sim} p(x_i|z_i) \quad i = 0, 1, \ldots, N$$

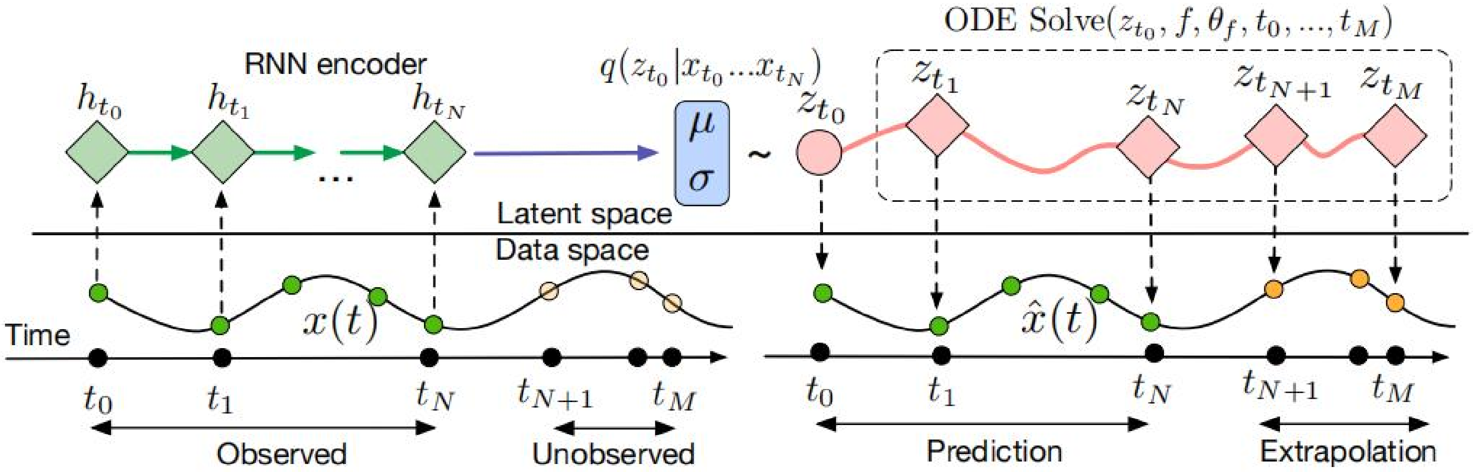
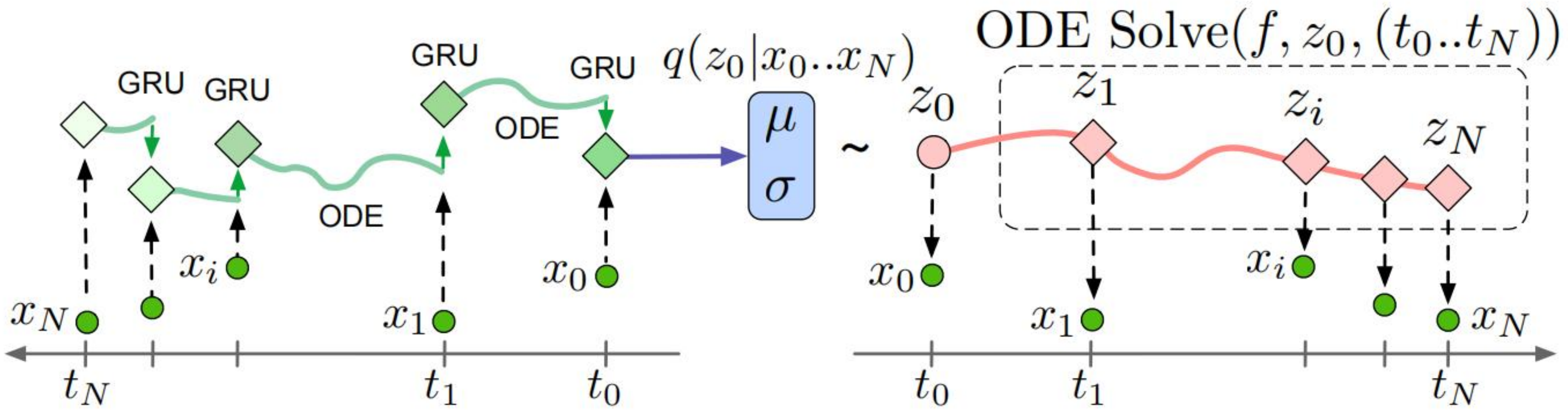


Figure 6: Computation graph of the latent ODE model.

Latent ODEs



$$q(z_0|\{x_i, t_i\}_{i=0}^{N}) = \mathcal{N}(\mu_{z_0}, \sigma_{z_0}) \quad \text{where} \quad \mu_{z_0}, \sigma_{z_0} = g(\text{ODE-RNN}_\phi(\{x_i, t_i\}_{i=0}^{N}))$$

Encoder-decoder models	Encoder	Decoder
Latent ODE (ODE enc.)	ODE-RNN	ODE
Latent ODE (RNN enc.)	RNN	ODE
RNN-VAE	RNN	RNN

泊松过程似然性

条件分布 $p(X|T) \longrightarrow p(X,T)$ 联合分布

$$p(\text{event at time } t|\,\mathbf{z}(t)) = \lambda(\mathbf{z}(t))$$

对数似然公式：

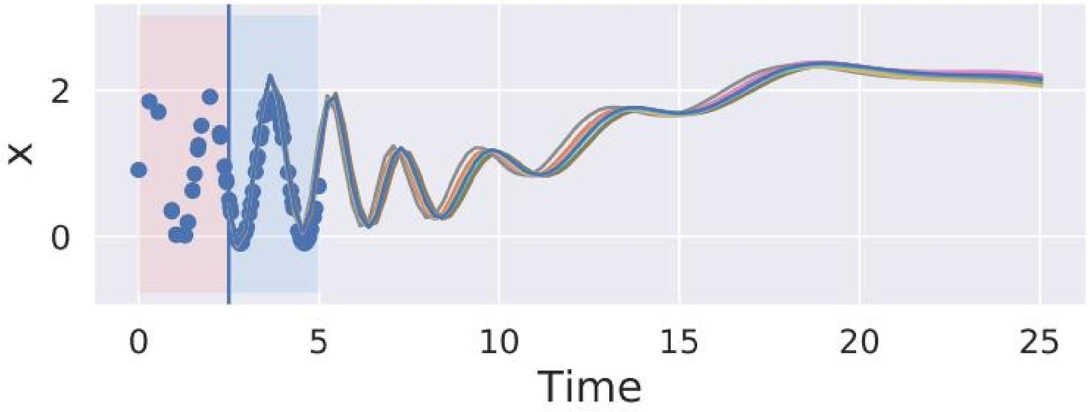
$$\log p(t_1, \ldots, t_N | t_{\text{start}}, t_{\text{end}}, \lambda(\cdot)) = \sum_{i=1}^{N} \log \lambda(t_i) - \int_{t_{\text{start}}}^{t_{\text{end}}} \lambda(t) dt$$

实验

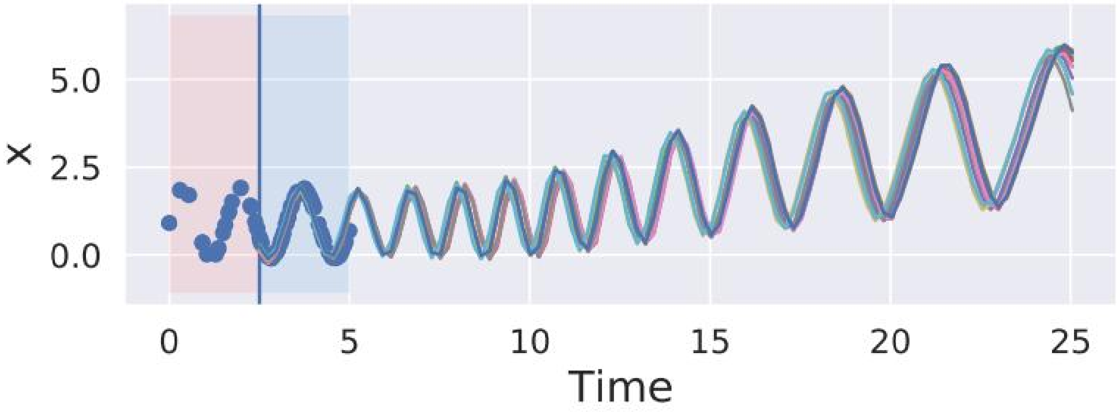
encoder

RNN VS ODE-RNN

(a) Latent ODE with RNN encoder



(b) Latent ODE with ODE-RNN encoder

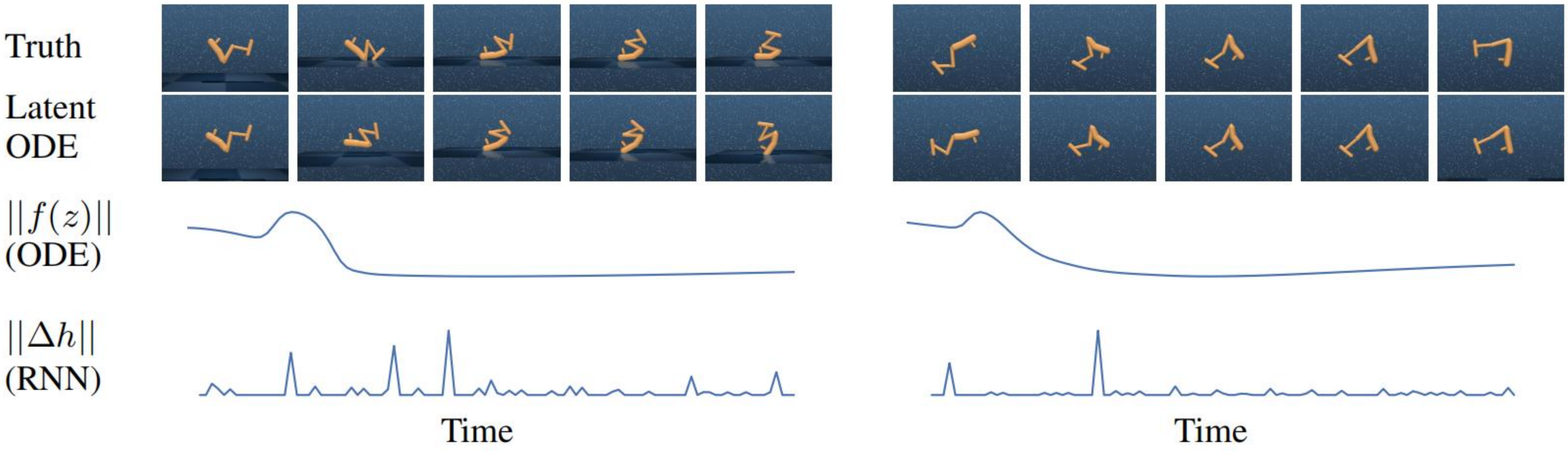


实验

Table 3: Test Mean Squared Error (MSE) ($\times 10^{-2}$) on the MuJoCo dataset.

	Model	Interpolation (% Observed Pts.) 10%	20%	30%	50%	Extrapolation (% Observed Pts.) 10%	20%	30%	50%
Autoreg	RNN Δ_t	2.454	1.714	1.250	0.785	**7.259**	**6.792**	**6.594**	30.571
	RNN GRU-D	1.968	1.421	1.134	0.748	38.130	20.041	13.049	**5.833**
	ODE-RNN (Ours)	**1.647**	**1.209**	**0.986**	**0.665**	13.508	31.950	15.465	26.463
Enc-Dec	RNN-VAE	6.514	6.408	6.305	6.100	2.378	2.135	2.021	1.782
	Latent ODE (RNN enc.)	2.477	0.578	2.768	0.447	1.663	1.653	1.485	1.377
	Latent ODE (ODE enc, ours)	**0.360**	**0.295**	**0.300**	**0.285**	**1.441**	**1.400**	**1.175**	**1.258**

实验



Latent ODE win!!!